

VibeAI

MULTI-PROVIDER
ORCHESTRATION SYSTEM
v4 · 2026-07

An orchestration system that makes ~12 free-tier AI models behave like one reliable engineering team — through verification infrastructure, not model size.

FRONTIER-QUALITY OUTPUT / ENGINEERED RESILIENCE

01 PROBLEM & SOLUTION

THE PROBLEM

Frontier models like Claude and GPT-4 sit behind paid access. Free-tier models are individually weak, and their providers are unreliable — they rate-limit hard, deprecate models overnight, and fail under load.

41% failure rate — OpenRouter under load

~20 req/day/model — Gemini free

8,000 TPM — Groq gpt-oss-120b

THE SOLUTION

Wrap the whole free-model fleet in verification infrastructure: a zero-token deterministic checker, a build-verification gate, automatic cross-provider failover, and autonomous escalation to any capable model before ever giving up.

Goal: match frontier quality

through engineering, not model strength

02 SYSTEM ARCHITECTURE

User

CLI · REST API · VS Code extension

Router Team

fast classification · local Ollama edge tier tried first

Free Manager Council

Plan → Draft → Critique → Refine → Polish · 5 models

Specialized Teams

Brain · Code · Design · Vision

Agent Loop

writes real files · runs builds & tests

Verifier Battery + Fix Cycles

deterministic checks · comparison-judged retries

CROSS-CUTTING

Model Registry

34 slots
10 providers

Circuit Breaker

parses live retry-delays
per-endpoint quota
memory

4-Tier Fallback

+ autonomous escalation
to any capable model

03 CORE CAPABILITIES

FEATURE 01

34-Slot Model Registry

10 free-tier providers, unified through cross-provider failover.

FEATURE 02

4-Tier Fallback + Escalation

A run keeps going even after 5 models fail in a row, in a fixed chain.

FEATURE 03

Deterministic Verifier Battery

Zero-token, zero-hallucination checks for defects models actually ship.

FEATURE 04

Mixture-of-Agents Core

Parallel proposers, an aggregator, and a self-consistency vote.

FEATURE 05

Adaptive Routing Memory

Learns which model wins per task category from real build/test outcomes.

FEATURE 06

Local Ollama Edge Tier

A free local 3B model handles quick classification, discovered live.

04 MEASURED RESULTS

7

DETERMINISTIC CHECKS IN THE ZERO-TOKEN VERIFIER BATTERY

172

AUTOMATED TESTS PASSING

34 / 10

REGISTRY ENTRIES ACROSS PROVIDERS

41%→0

OPENROUTER'S RAW LOAD FAILURE RATE, ABSORBED BY THE RESILIENCE LAYER

4+∞

FALLBACK TIERS, THEN UNLIMITED ESCALATION DEPTH

0 / 67

PRIMARY-MODEL FAILURES ACROSS LOGGED LIVE CALLS

05 NOVEL TECHNIQUES

Mixture-of-Agents (Wang et al., 2024)

CeP0-scoped comparison judging

SONA-style routing memory

Progressive tool disclosure

06 DIFFERENTIATORS

- Radical honesty — reversed its own fabricated "31-model" branding; every claim is live-verified.
- Verification > model strength as the core design philosophy.
- Studies real competitor **source code** and ports only what transfers.
- Bounded, interpretable learning instead of unsafe self-improvement loops.